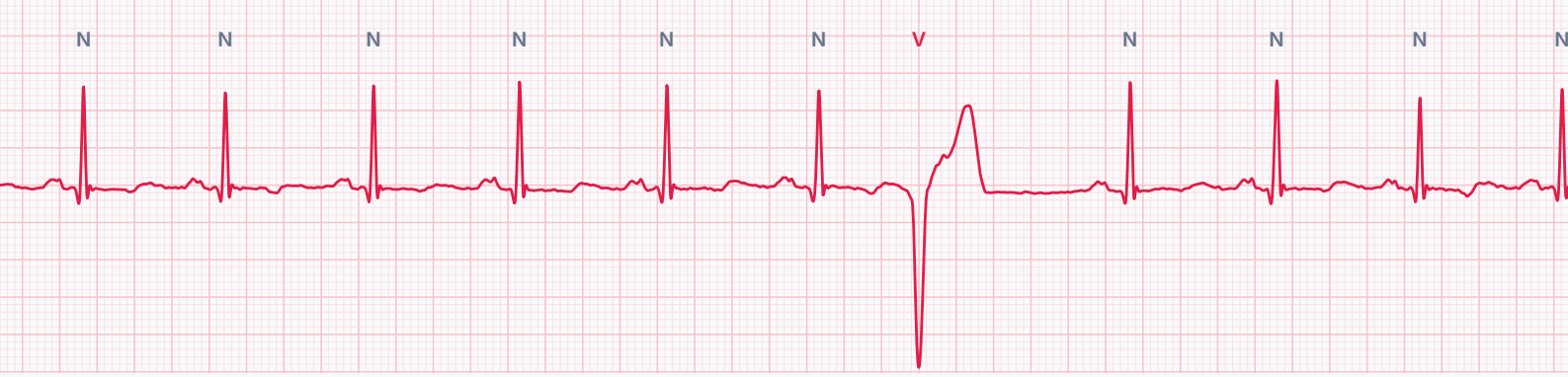


ECG Arrhythmia · project report

MIT-BIH Arrhythmia Database | ANSI/AAMI EC57 | Inter-patient DS1 / DS2 | Built with orchestrated AI agents

Preliminary results

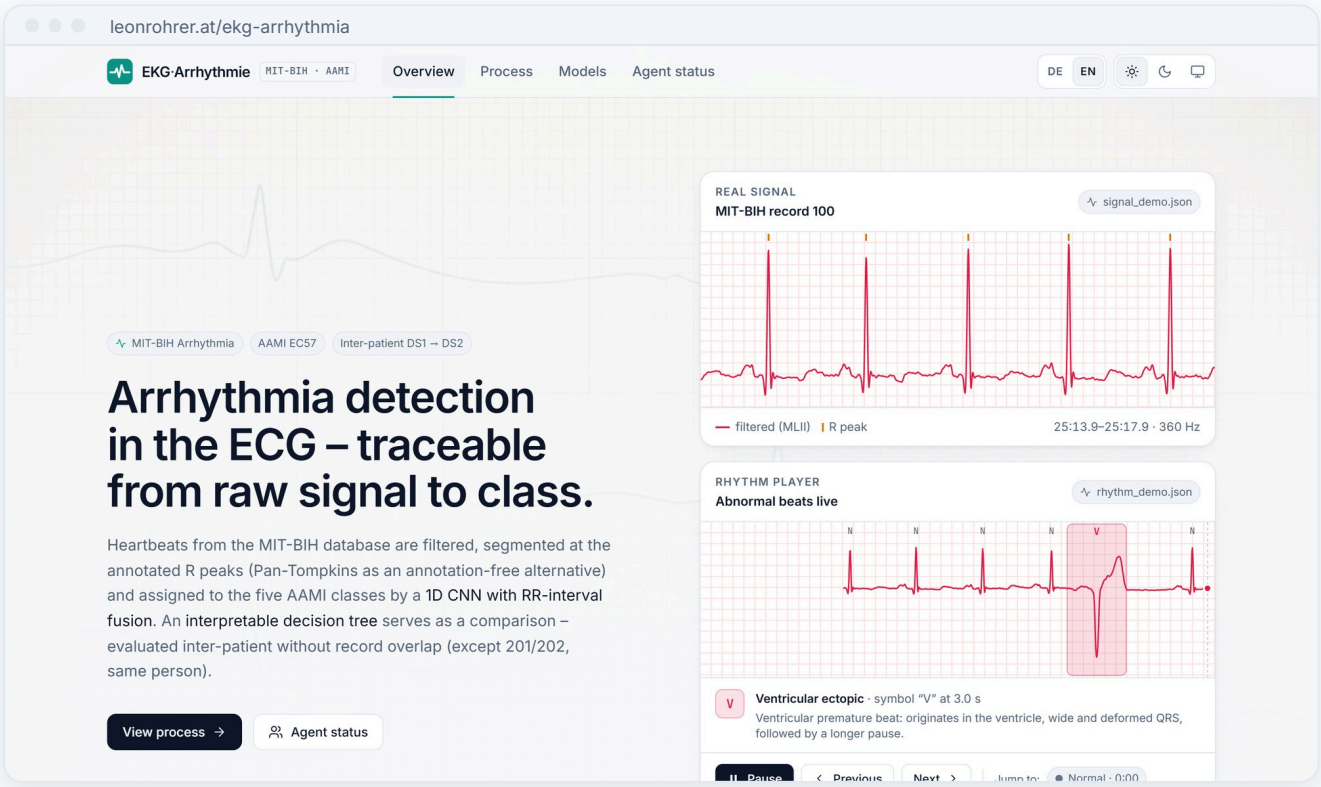
Arrhythmia detection in the ECG — from raw signal to beat class

A complete, reproducible pipeline that sorts every heartbeat of a long-term ECG into the five AAMI classes, with a compact 1D convolutional network, an interpretable decision tree and an honest inter-patient evaluation on patients the models have never seen.

AUTHOR
Leon Rohrer

DATE
3 November 2026

INTERACTIVE VERSION
leonrohrer.at/ekg-arrhythmia



SUMMARY

Finding the rare beats that matter

Every heartbeat of an ECG recording is assigned to one of the five classes of the ANSI/AAMI EC57 standard. The beats that matter clinically — premature beats from the atria (S) or from the ventricles (V) — are rare: 89.0% of all test beats are normal. This project builds the complete chain from the raw signal to the class and evaluates it the strict way, on the records of 22 patients that the models never saw during development.

Problem

Ectopic beats are clinically relevant but hidden among thousands of normal beats. Many published results mix beats of the same patient into training and test data, which inflates the scores far beyond what a new patient would see.

Approach

MIT-BIH Arrhythmia Database, split inter-patient into DS1 (training) and DS2 (test) after de Chazal et al. (2004). Filtering, Pan-Tompkins R-peak detection, one window per beat plus RR intervals; a compact 1D CNN (14,749 parameters) and an interpretable decision tree.

Built with AI agents

A coordinating “mother agent” split the work into 18 modules for up to 16 parallel sub-agents, held together by binding interface contracts and a lock-safe status protocol. Goals, rules and final checks stayed with the human.

Key result: 1D CNN on the unseen test patients (DS2)

S SUPRAVENTRICULAR BEATS

69.0% sensitivity — **40.2%** PPV

1,837 S beats in the test set

V VENTRICULAR BEATS

79.2% sensitivity — **93.5%** PPV

3,220 V beats in the test set

MACRO-F1

0.58 N, S, V, F — **0.46** all five classes

Accuracy 89.6% is only a side note: answering “normal” every time would already reach 89.0%.

Sensitivity is the share of the true beats of a class that the model found; the **positive predictive value (PPV)** is the share of the model's alarms for that class that were right. For orientation: de Chazal et al. (2004) reported Se 75.9% / PPV 38.5% for S and Se 77.7% / PPV 81.9% for V on the same split (details in chapter 5). The interpretable decision tree reaches a macro-F1 of 0.36 (N, S, V, F).

Read with care

- **Preliminary version:** the metrics come from a development run (`metrics.json` of 2026-11-02) before the final, one-time evaluation on DS2 and will be replaced by the final results.
- The test split DS2 was evaluated twice during development (documented). Since then it is locked: further decisions use cross-validation on DS1 only, and DS2 is evaluated once for the final result.
- F and Q are practically not recognized (F1 0.00 and 0.00): DS1 has too few and too one-sided examples of them.
- This is a research and portfolio project, not a medical device; it must not be used for diagnosis.

Interactive version with all steps and live charts: https://leonrohrer.at/ekg-arrhythmia/ · Code: https://github.com/LeonTheGoatRohrer/ekg-arrhythmia

1 · BACKGROUND

What the ECG shows — and why evaluation is tricky

The ECG in one minute

The heart is a muscle that contracts about once per second, each time triggered by an electrical impulse from the **sinus node**, its built-in pacemaker. Electrodes on the skin pick up this activity; the voltage over time is the **electrocardiogram (ECG)**.

A normal beat has a typical shape: the small **P wave** (the atria contract), the sharp **QRS complex** (the large ventricles contract) with its tallest spike, the **R peak**, and the broader **T wave** (the ventricles recover). The time between two R peaks is the **RR interval** — at 60 beats per minute exactly one second.

An **arrhythmia** is a disturbance of this rhythm: a beat comes too early, from the wrong place or with the wrong shape. Many are harmless, some are dangerous.

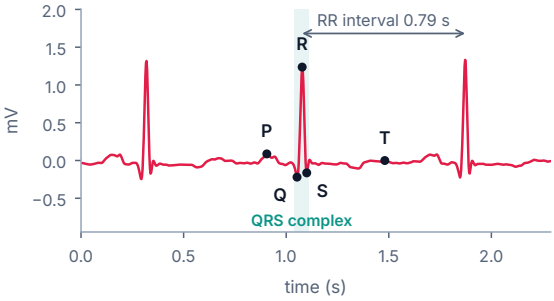


Figure 1. Three normal beats from MIT-BIH record 100 (filtered), with the waves of the middle beat and one RR interval marked.

Five beat classes (ANSI/AAMI EC57)

The MIT-BIH annotations distinguish more than a dozen beat types. To make results comparable, the **ANSI/AAMI EC57** standard groups them into five classes (annotations that do not mark a beat are ignored). S and V carry most of the clinical value — frequent supraventricular ectopic beats can precede atrial fibrillation, frequent or coupled ventricular ones point to structural heart disease — and they are also the rare ones, which makes them hard for algorithms.

Class	Meaning	MIT-BIH symbols	Typical cue
N	Normal and bundle branch block beats	`N, L, R, e, j`	regular timing, normal shape
S	Supraventricular ectopic (from the atria)	`A, a, J, S`	too early; shape often almost normal
V	Ventricular ectopic (from the ventricles)	`V, E`	wide, deformed QRS; pause afterwards
F	Fusion of a normal and a ventricular beat	`F`	shape in between N and V
Q	Paced and unclassifiable beats	`/, f, Q`	almost absent without paced records

Why inter-patient instead of intra-patient

Beats from the same person look very much alike. If all beats are shuffled and split at random (*intra-patient*), the test set contains near-copies of beats the model already knows: it learns "what does patient X look like" instead of "what does a ventricular beat look like". Many published accuracies around 99% rest on such splits — but a new patient is, by definition, unseen. This project therefore follows the **inter-patient** protocol of de Chazal et al. (2004): two disjoint sets of 22 records, DS1 for everything during development and DS2 only for testing.

Intra-patient

train

test

Same patients (colors) in both — the model recognizes people, the score is inflated.

Inter-patient

train

test

Unseen patients — a realistic estimate for a new person.

2 · DATA

MIT-BIH: 44 usable records, 22 + 22 for a fair test

The **MIT-BIH Arrhythmia Database** (Moody & Mark, 2001; distributed by PhysioNet) is the reference dataset of arrhythmia research: 48 half-hour excerpts of two-channel ambulatory ECGs from 47 subjects, recorded at Beth Israel Hospital in Boston, sampled at 360 Hz with 11-bit resolution. At least two cardiologists annotated every single beat independently; disagreements were resolved by consensus. The project uses lead `MLII`, in which the R peaks are particularly clear.

- **Excluded:** the four records with a pacemaker (102, 104, 107, 217), as recommended by AAMI — paced beats have their own shape. 44 records remain.
- **Same person:** records 201 and 202 come from the same subject but sit in different sets — a known, small imprecision of the standard split, accepted for comparability with the literature.

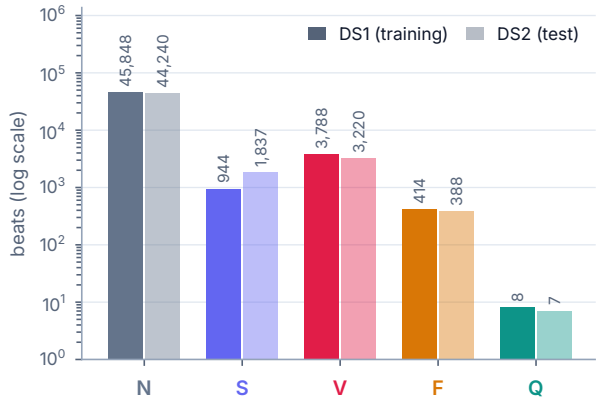


Figure 2. Beats per class and split (dataset_stats.json). The axis is logarithmic: N outnumbers S, F and Q by two to four orders of magnitude.

DS1 · training and validation · 22 records

`101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230`

DS2 · test only · 22 records

`100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234`

Class distribution

Class	DS1 beats	share	DS2 beats	share
N normal	45,848	89.9%	44,240	89.0%
S supraventricular ectopic	944	1.9%	1,837	3.7%
V ventricular ectopic	3,788	7.4%	3,220	6.5%
F fusion	414	0.8%	388	0.8%
Q unclassifiable	8	0.0%	7	0.0%
Total	**51,002**		**49,692**	

Why this matters later

N makes up 89.9% of DS1 and 89.0% of DS2. DS1 contains only 414 fusion (F) beats — about 90% of them from a single record, 208 — and 8 Q beats. Classes this small and this concentrated can hardly be learned in a way that transfers to new patients, and every metric on them is very noisy.

3 · PIPELINE

From raw signal to beat: the pipeline step by step

Seven stations turn a half-hour recording into one class label per heartbeat. All signal figures in this chapter show real data from MIT-BIH record 100 (part of the test set DS2; used here only for illustration, never for training).



3.1 Filtering without shifting the beats

A real ECG is never clean. **Baseline wander** — a slow drift caused mainly by breathing and electrode movement — lies well below 1 Hz; **power-line interference** sits at 60 Hz in this US database; **muscle noise** is broadband and fast. The information that matters for beat classification lies roughly between 0.5 and 40 Hz.

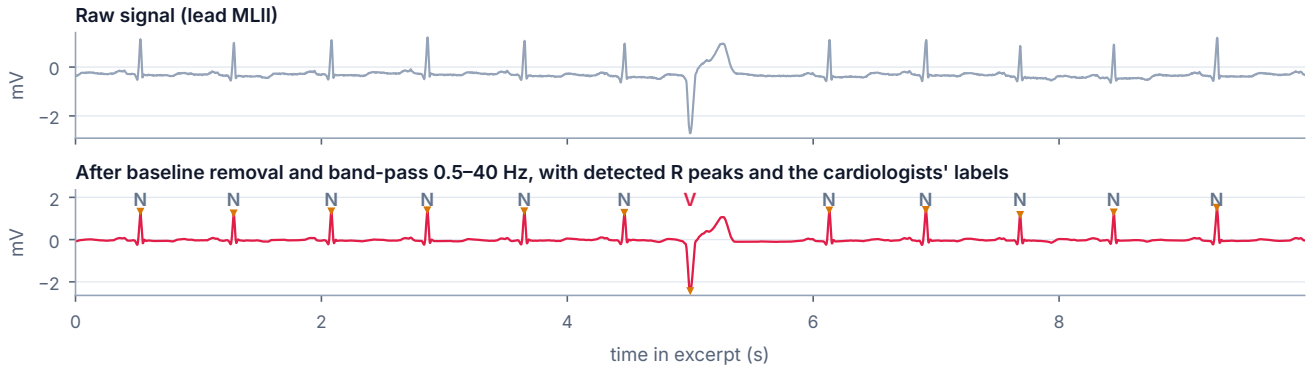


Figure 3. A 10-second excerpt of record 100 before and after filtering. Letters above the beats are the cardiologists' classes; triangles mark the R peaks found by the detector.

Stage 1 – baseline removal (after de Chazal et al., 2004): a 200 ms median filter removes QRS complexes and P waves, a second one with 600 ms also removes the T waves; the remaining baseline estimate is subtracted. Median filters are robust against the steep QRS spikes and do not smear them like a moving average.

Stage 2 – band-pass 0.5–40 Hz: a Butterworth filter of order 4 removes the remaining drift, muscle noise and mains hum. Filtering forward and backward (`sosfiltfilt`) cancels the phase shift, so the R peak stays exactly in place — important because the next steps cut fixed windows around it.

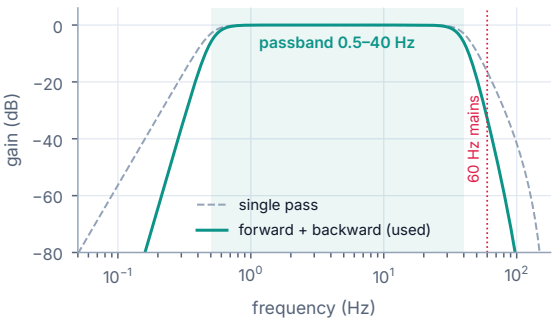


Figure 4. Frequency response of the band-pass (filter_response.json). Filtering forward and backward doubles the attenuation outside the passband.

3.2 Finding every beat: Pan-Tompkins R-peak detection

Before beats can be classified, each one has to be found. The classic algorithm of **Pan & Tompkins (1985)** turns the ECG into a signal in which every QRS complex becomes one clear bump: a 5–15 Hz band-pass emphasizes the QRS and suppresses P and T waves, the derivative highlights steep slopes, squaring makes everything positive and amplifies large slopes, and a 150 ms moving-window integration smooths this into an envelope about as wide as a QRS complex. Adaptive thresholds then decide what is a beat, with a 200 ms refractory period, a T-wave check and a search-back when a beat seems to be missing.

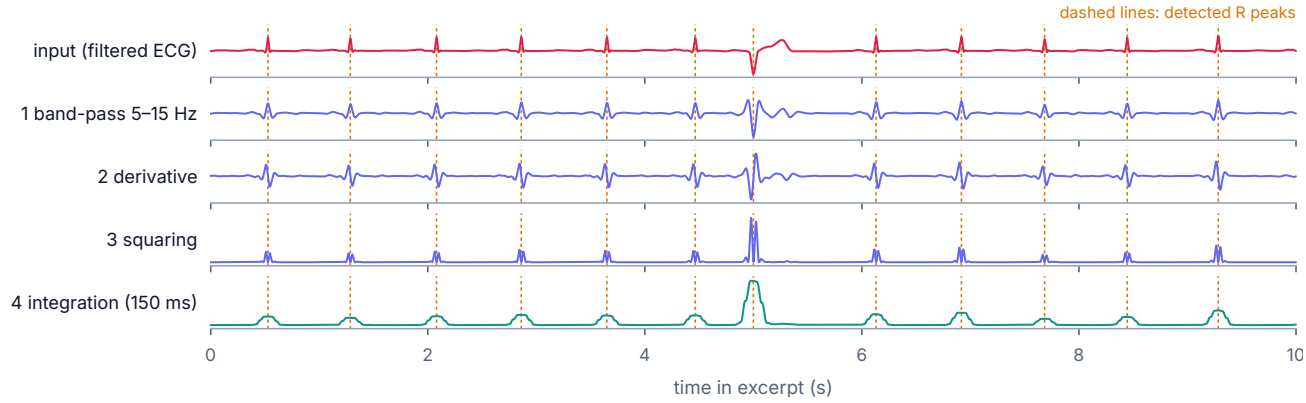


Figure 5. The intermediate stages of the detector on the same excerpt (each row scaled to its maximum).

> **Check on record 100:** 2,273 of 2,273 annotated beats found, 0 false detections (tolerance ±150 ms) — a single, rather clean record, not a database benchmark. For training and evaluation, the models use the cardiologists' reference beat positions, as in de Chazal et al. (2004), so that detection errors do not mix into the classification result.

3.3 One window per beat

Around every R peak, a window from **250 ms before** to **450 ms after** the peak is cut out — 252 samples at 360 Hz, R always at the same index. The asymmetric window contains the P wave before the QRS and the longer T wave after it; both carry diagnostic information. Each window gets its class from the cardiologists' annotation and is scaled to mean 0 and standard deviation 1, so the model learns the *shape* rather than the amplitude, which varies with electrode position and body build.

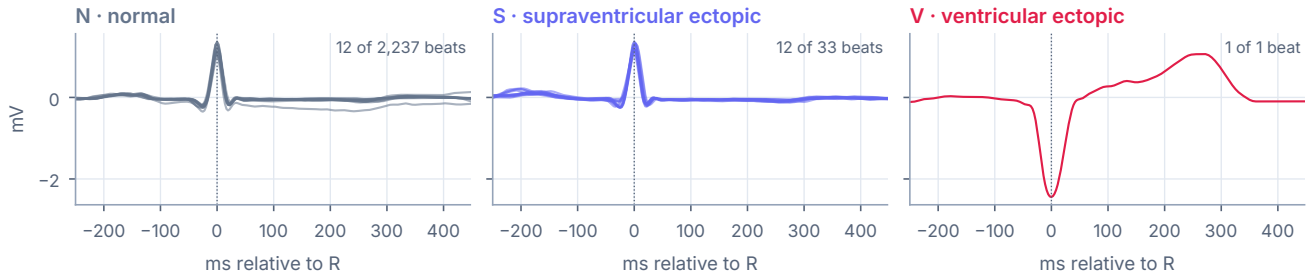


Figure 6. Real beat windows from record 100 (segmentation_demo.json), before z-normalization. The V beat is wide and deformed; the S beats look almost normal. Classes F and Q do not occur in record 100.

3.4 Rhythm context: the timing gives an early beat away

A window shows the shape but not the timing. Four RR features add it: `pre_rr` (to the previous beat), `post_rr` (to the next), `local_rr` (mean of the last up to ten intervals) and the ratio `pre_rr / local_rr`. Ectopic beats come *early*, so the ratio drops well below 1, and a ventricular beat is often followed by a compensatory pause. For S beats, whose shape is often nearly normal, prematurity is frequently the decisive cue; the ratio also keeps it comparable between people with different heart rates.

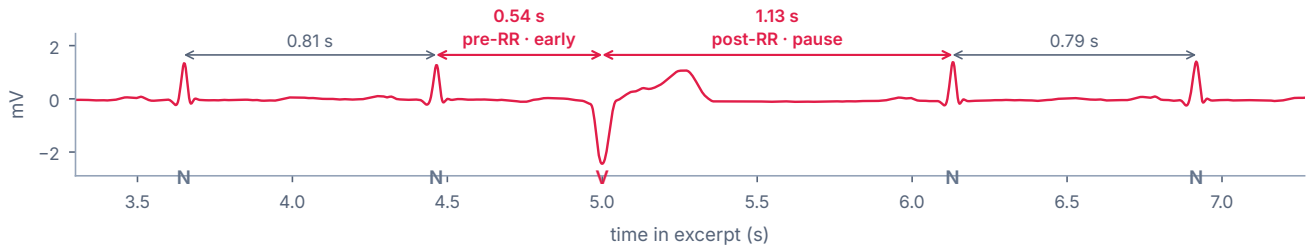


Figure 7. RR intervals around the V beat of the excerpt: pre-RR 0.54 s against a local rhythm of about 0.75 s (ratio ≈ 0.72), followed by a 1.13 s pause.

Two models: a compact CNN and an interpretable tree

4.1 A compact 1D convolutional network

A **1D convolutional neural network** slides small learned filters across the beat window. Each filter is a pattern detector that responds where the signal resembles its shape — a steep slope, a wide QRS, an inverted wave. Unlike the features of the decision tree, these patterns are not defined by hand but learned from the data. Three convolution blocks (convolution, batch normalization, ReLU, max-pooling) halve the time resolution step by step, so deeper layers see larger parts of the beat. A global average then condenses each channel into one number. In parallel, the four RR features pass through a small dense layer. Both branches are concatenated and mapped by a small classification head to one score per AAMI class — the network judges *shape and timing together*. With 14,749 trainable parameters it is deliberately small, leaving little room to memorize the 18 training records, and the whole training run took about 1.5 minutes on a CPU.

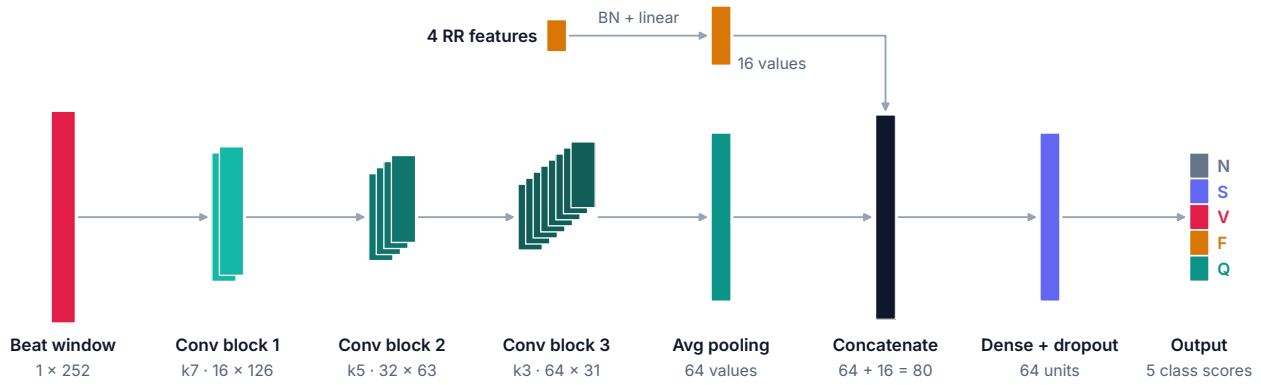


Figure 8. ECGNet1D as drawn from architecture.json: the time axis shrinks from 252 samples while the number of learned channels grows (k = kernel size); the RR branch joins before the classification head.

Stage	Layers	Output shape	Parameters
Beat window (input)	Input	1 × 252	0
RR features (input)	Input	4	0
Conv block 1	Conv1d · BatchNorm1d · ReLU · MaxPool1d	16 × 126	160
Conv block 2	Conv1d · BatchNorm1d · ReLU · MaxPool1d	32 × 63	2,656
Conv block 3	Conv1d · BatchNorm1d · ReLU · MaxPool1d	64 × 31	6,336
Global average pooling	AdaptiveAvgPool1d · Flatten	64	0
RR branch	BatchNorm1d · Linear · ReLU	16	88
Concatenate	Concat	80	0
Dense layer + dropout	Linear · ReLU · Dropout	64	5,184
Output layer	Linear	5	325
Total			**14,749**

4.2 Training against the imbalance

Weighted loss. The cross-entropy loss weights every class inversely to its frequency in the training data, capped at 50× the weight of the most common class. A missed S beat therefore costs much more than a missed N beat; without this, the network would largely ignore the rare classes.

Optimizer. Adam with learning rate 0.001, mini-batches of 256 beats, up to 20 epochs, fixed random seed 0 for reproducibility.

Validation inside DS1 only. Records 114, 119, 122, 201 of DS1 are held out for early stopping (patience 5 epochs). DS2 is used neither for choosing settings nor for the stopping point — otherwise test information would leak into the model. Records that hold a large share of a rare class stay in training: record 208 alone holds about 90% of all DS1 fusion beats. In the run reported here, training stopped after 7 epochs and the weights of epoch 2 were kept.



Figure 9. Training history (metrics.json). The dotted line marks the epoch whose weights were kept.

4.3 An interpretable decision tree for comparison

The second model is a decision tree of depth 4 with 16 leaves, trained on 51,002 DS1 beats with balanced class weights. Instead of raw samples it receives 11 hand-crafted features that a cardiologist would recognize. Every path from the root to a leaf is a readable if-then rule — the price is far less expressive power than the CNN.

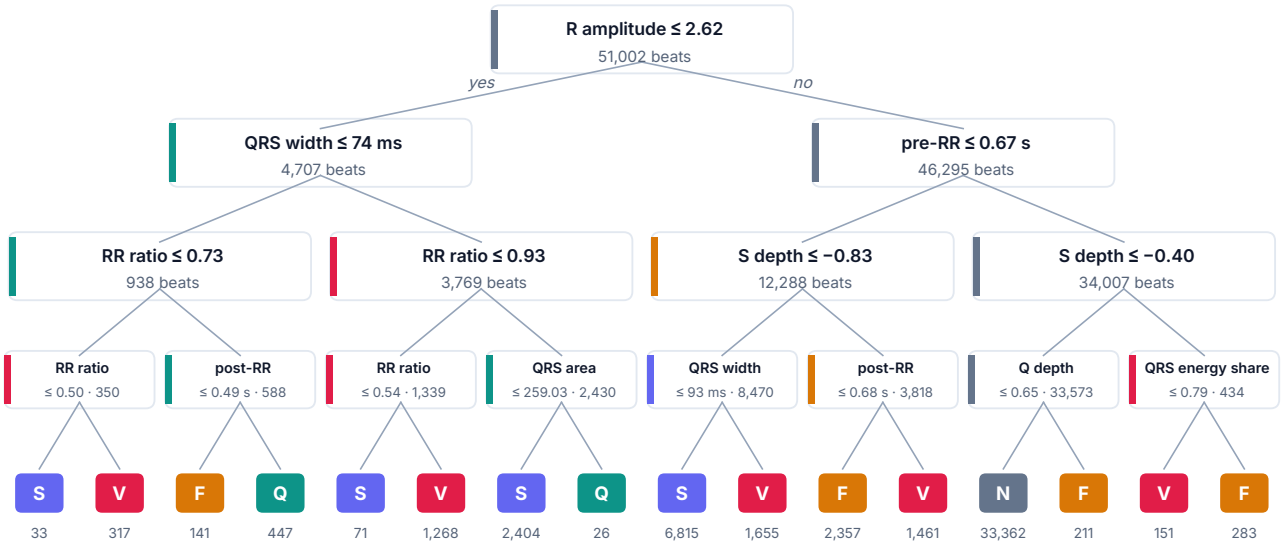


Figure 10. The learned tree (tree.json). Each box tests one feature; the left branch is taken when the condition holds. Colored bars and leaves show the class the tree would predict (with balanced class weights); numbers are training beats reaching the node. Amplitudes are in units of the normalized beat.

- `qrs_width_ms` — duration of the QRS complex
- `r_amplitude` — height of the R peak (normalized beat)
- `q_depth` — deepest point just before R
- `s_depth` — deepest point just after R
- `qrs_area` — area under the QRS complex
- `t_amplitude` — height of the T wave
- `beat_energy` — share of the beat's energy inside the QRS
- `pre_rr` — interval to the previous beat
- `post_rr` — interval to the next beat
- `local_rr` — mean of the last up to 10 RR intervals
- `rr_ratio` — pre-RR / local RR (prematurity)

5 · EVALUATION

Results on unseen patients (DS2)

All numbers in this chapter come from `metrics.json` (generated 2026-11-02): both models evaluated on the 22 test records of DS2 — 49,692 beats from patients the models never saw.

Sensitivity (Se) = TP / (TP + FN): how many of the true beats of a class were found.

Positive predictive value (PPV, +P) = TP / (TP + FP): how many of the model's alarms for a class were right. **FPR** = FP / (FP + TN): false alarms among all other beats.

F1: harmonic mean of Se and PPV, high only if both are high. **Macro-F1** averages F1 over classes, so rare classes count as much as N.

Per-class results on DS2

		Sensitivity		PPV		F1	
Class	Test beats	CNN	Tree	CNN	Tree	CNN	Tree
N	44,240	**92.0%**	60.0%	**97.5%**	95.0%	**0.95**	0.74
S	1,837	**69.0%**	7.1%	**40.2%**	1.9%	**0.51**	0.03
V	3,220	79.2%	**82.4%**	**93.5%**	54.6%	**0.86**	0.66
F	388	1.0%	**1.8%**	**0.2%**	0.1%	**0.00**	0.00
Q	7	0.0%	**42.9%**	0.0%	**0.2%**	0.00	**0.00**
Macro-F1 (N, S, V, F)						**0.58**	**0.36**
Macro-F1 (all five)						0.46	0.29
Accuracy (supplementary)						89.6%	59.0%

The better of the two models is set in bold. Accuracy is listed only for completeness: with 89.0% normal beats it mostly measures the majority class.

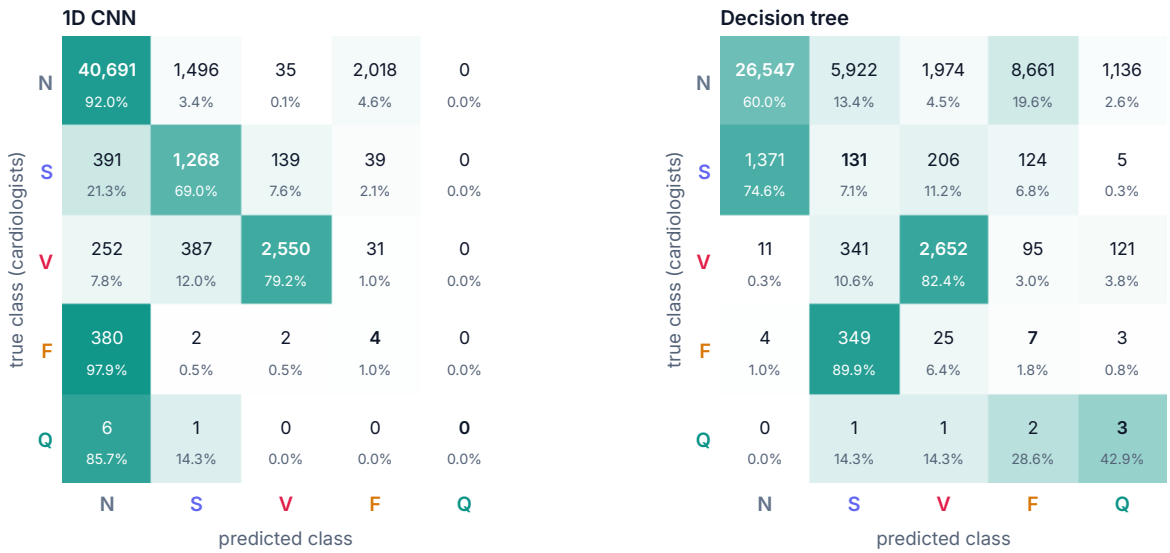


Figure 11. Confusion matrices on DS2: rows are the cardiologists' classes, columns the predictions; each row is colored by the share of its beats (percentages below the counts).

Reading the CNN's matrix

The three largest confusions are:

- 2,018 true N beats predicted as F (4.6% of all N beats)
- 1,496 true N beats predicted as S (3.4% of all N beats)
- 391 true S beats predicted as N (21.3% of all S beats)

The decision tree stays clearly below the CNN (macro-F1 0.36 vs. 0.58 on N, S, V, F): a handful of threshold rules on hand-made features is easy to read, but cannot capture the subtle shape differences the network learns.

Where the errors come from

Missed S beats are mostly called N (391 of 569 misses); false S alarms come mostly from true N beats (1,496 of 1,886). Missed V beats are mostly called S (387 of 670 misses); false V alarms come mostly from true S beats (139 of 176). Missed F beats are mostly called N (380 of 384 misses); false F alarms come mostly from true N beats (2,018 of 2,088).

Comparison with the literature baseline

De Chazal, O'Dwyer & Reilly (2004) introduced the DS1/DS2 protocol and reported, for their selected configuration on DS2, the values below (from the paper's abstract). It is the natural reference because it uses the same split and the same reference beat positions. A difference to keep in mind: their linear-discriminant classifier used hand-crafted morphology and interval features from *both* ECG leads, whereas this project uses one lead.

Class	Model	Se	+P	FPR
S	1D CNN (this project)	69.0%	40.2%	3.9%
S	Decision tree (this project)	7.1%	1.9%	13.8%
S	de Chazal et al. 2004	75.9%	38.5%	4.7%
V	1D CNN (this project)	79.2%	93.6%	0.4%
V	Decision tree (this project)	82.4%	54.9%	4.7%
V	de Chazal et al. 2004	77.7%	81.9%	1.2%

Values for this project are gross statistics (all test beats pooled), as in the paper; VEB: true F beats classified as V are not counted as V false positives.

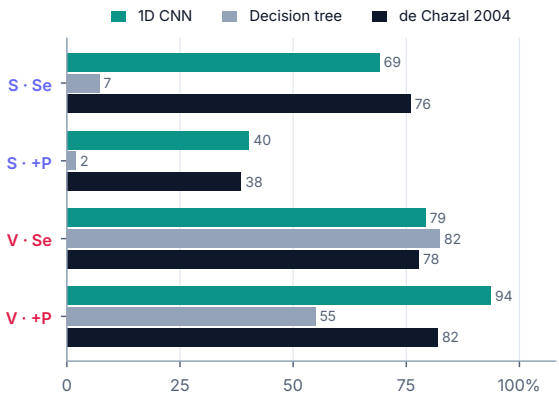


Figure 12. Sensitivity (Se) and positive predictivity (+P) for S and V beats, in percent.

6 · LIMITATIONS

Limitations and honest methodology

A result is only as useful as its honest limits. These are the ones that matter most here.

The test set was looked at more than once

DS2 should ideally be evaluated exactly once. In this project it was evaluated twice before the final run (run 1: validation records 114, 122, 201, 208, macro-F1 over five classes 0.447; run 2: validation records 114, 119, 122, 201, macro-F1 over five classes 0.463; see `docs/EXPERIMENTS.md`). Between the runs, the choice of validation records was changed: record 208 holds about 90% of all DS1 fusion beats and was moved from validation back into training. The reason was a property of DS1, but the change was made after a DS2 result was known, so a small optimistic bias cannot be ruled out. Since then the rule is strict: DS2 is locked, every decision is made by record-grouped cross-validation on DS1 with a selection criterion fixed in advance, and DS2 is evaluated once at the very end — and reported whatever the outcome.

F and Q are practically not learnable here

DS1 contains 414 fusion beats, about 90% of them from one patient (record 208), and only 8 Q beats; DS2 has 7 Q beats. The CNN finds 1.0% of the 388 F beats in DS2 and 0.0% of the Q beats; its F alarms are right in 0.2% of cases. With so few, one-sided examples, a model cannot learn a pattern that transfers to new patients — larger and more diverse datasets would be needed.

Supraventricular beats (S)

The CNN finds 69.0% of all S beats, but only 40.2% of its S alarms are correct. S beats often look like normal beats and are mainly recognized by their timing, which also varies naturally in some patients.

Reference beat positions

Classification is evaluated on the cardiologists' beat positions, the usual protocol for comparability. A fully automatic system would also inherit the detector's errors, so its results would be somewhat lower; the detector itself was only checked in detail on single records.

Small, old, single-source data

44 usable records from one hospital, recorded in the late 1970s with Holter devices, one lead used. All results in this report come from MIT-BIH; it contains no external validation on other databases or modern wearables.

Run-to-run variation

The reported model is a single training run (seed 0). Cross-validation inside DS1 showed that results move noticeably with the random seed and the fold split, so differences of a few points between variants should not be over-interpreted.

Not a medical device

This is a research and portfolio project. It has no clinical validation and no regulatory approval, and it must not be used to diagnose or treat anyone.

What was done to keep the evaluation honest

- Strict inter-patient split; no record in both DS1 and DS2 (the only shared person, 201/202, is disclosed).
- Per-class sensitivity, PPV and F1 as the main metrics; accuracy only as a side note.
- Model selection, early stopping and every design decision on DS1 only; a selection rule (improvement of at least one standard deviation in cross-validation, without losing N sensitivity) was fixed *before* the experiments and is applied even when a promising variant narrowly fails it.
- All experiments logged with date, configuration and result in `docs/EXPERIMENTS.md`; the web interface shows empty states instead of placeholder numbers whenever a result does not exist.

7 · PROCESS

How it was built: orchestrated AI agents

The project was planned and implemented with orchestrated AI agents. The point was not to let a model “write some code”, but to run software development like a small team: clear ownership, binding contracts, a shared status file and independent review.

Human (Leon Rohrer) — sets the goal and the ground rules (inter-patient evaluation, per-class metrics, no invented numbers), approves plans and anything that touches the test set or goes public, reviews results and texts.

↓

Mother agent (orchestrator) — writes the master plan, cuts the work into 18 modules, reads `project_state.json` before every wave and starts only modules whose dependencies are done (at most 16 sub-agents at once).

↓

Wave 1 · 13 agents in parallel on disjoint files

Data pipeline · 4	ML model · 4	Web & UI · 5
loader, filters, R peaks, segmentation	1D CNN, decision tree, training, evaluation	API, shell, diagrams, charts, content

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Wave 2 · integration — end-to-end pipeline CLI, then the web UI on real data.

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Wave 3 · independent review — three reviewer agents with different angles (ML correctness and leakage, tests, UI, honesty of claims); 17 findings, all reproduced and fixed by a fix agent.

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Wave 4 · full training DS1 → DS2 — blocked in the cloud sandbox (no access to PhysioNet), completed locally on Windows.

Status protocol
Agents never edit the status file directly. They report through `scripts/state.py`, which locks the file:
`start` · `done` (summary, decisions, files, tests) · `fail` (log excerpt → corrected restart) · `log` (deviation from a contract)

Interface contracts
`INTERFACES.md` fixes every function signature and data format up front, so modules can be built in parallel against each other.

Ownership
Each module owns its files exclusively; nobody edits another agent's files.

Tests
Every module ships with its own tests; the suite now has 214 tests. Agents report test commands and results with `done`.

Why this is more than a gimmick

Parallel agents only work if the problem is decomposed well. The contracts in `INTERFACES.md` made 13 agents productive at the same time without stepping on each other; the lock-safe status file made progress, decisions and failures traceable; and a separate review wave caught real bugs (for example a too-narrow QRS width measurement and uncapped class weights) before any result was published. The human role shifted from typing code to **specifying, deciding and verifying** — the skills that matter when automating real work.

Tech stack

Signal & ML	Python · NumPy, SciPy, wfdb, PyTorch, scikit-learn (python 3.13.5, numpy 2.5.4, scipy 1.18.1, torch 2.15.0, scikit-learn 1.9.1, wfdb 4.3.1)
Pipeline & API	command-line pipeline (`python -m ekg.pipeline`), FastAPI server, Makefile
Web interface	Vite, Tailwind CSS v4, Mermaid diagrams, hand-written SVG charts; German and English; hosted at leonrohrer.at
Quality	**214 automated tests** (counted with `pytest --collect-only` when this report was built); browser checks of the UI with Playwright; reproducible versions in `constraints.txt`
This report	generated by `docs/report/build_report.py` from the same JSON exports (matplotlib, HTML/CSS, Chromium)

EXPLORE IT YOURSELF

Every step, interactive

The web version shows the pipeline with live charts on real data, the rhythm player, both models and the agent status — in English and German.

leonrohrer.at/ekg-arrhythmia/

Live page	https://leonrohrer.at/ekg-arrhythmia/
Source code	github.com/LeonTheGoatRohrer/ekg-arrhythmia
Author	Leon Rohrer

References

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